Rahul Reddy Talatala

rahul.talatala@gmail.com | 716-939-5940 | rahult.dev | linkedin.com/in/rahul-reddy-t | github.com/rahult18

Summary

ML Engineer with 3 years of experience building agentic AI systems, recommendation pipelines, and LLM-powered infrastructure. Automated 60% of network triage at Verizon generating $2.46M annual cost avoidance across multi-agent telecom systems, and reduced infrastructure debugging time 60% at Apple through fine-tuned LLM diagnostics and GPU observability pipelines.

Skills

Agentic & ML: LangGraph, LangChain, Autogen, CrewAI, PyTorch, TensorFlow, HuggingFace, Google ADK, MCP, A2A
Foundation Models: GPT-4, Claude, Gemini, Llama, Mistral, Qwen
RAG & Knowledge Systems: Hybrid RAG, LlamaIndex, Neo4j, Elastic Vector DB, pgvector, Cohere Rerank, Semantic Search
MLOps & LLMOps: Ray, ONNX, MLflow, NVIDIA Triton, LangSmith, LangFuse, Kubeflow, W&B, Axolotl
Data Engineering: Spark, Airflow, Kafka, dbt, Snowflake, TimescaleDB, ETL/ELT, Data Warehousing, Data Modeling
Cloud & Deployment: AWS, GCP, Azure, Kubernetes, Terraform, Docker, Run:AI, ArgoCD, Github Actions, CI/CD
Programming: Python, SQL, Java, TypeScript/JavaScript, numpy, pandas, scikit-learn, FastAPI, Node.js, React

Experience

GenAI Engineer — Aug 2025 – Present
Infinite Computer Solutions, Client: Verizon — *Dallas, TX*

- Generated $2.46M annual cost avoidance as measured by 47,068 engineering hours reclaimed, by architecting the DS3M Disconnect Agent, a 3-agent LangGraph system covering Zero Fill, Last Customer Rider, and Data Integrity Clean Up scenarios that ingests real-time data from 9 network inventory and billing systems to identify unused circuits and auto-issue disconnect orders to LECs with minimal human intervention.
- Reduced manual fiber build planning effort by 65% as measured by cluster assignment SLAs, by building the FiOS Clustering Agent that runs geospatial DBSCAN clustering to group addresses into optimized build zones, then coordinates 6 specialized sub-agents (Copper Reclamation, Hub Capacity, Fiber Route, Copper Route, Cost, and Ranking) to produce a prioritized, execution-ready plan with hub mappings and cost summaries.
- Maintained 100% reporting accuracy across 14 revenue dashboards serving 250+ engineers, by building the Copper Recycling DAG Monitoring Agent that continuously watches 299 GCP Airflow pipelines, classifies failure types using pattern matching and heuristic rules, and auto-triggers targeted recovery workflows without manual intervention.
- Cut Mean Time To Detect for wireless cell site outages by 40%, by building the Out of Service Triage Agent that pulls diagnostic data from AMOS, ENM, and USM via REST APIs, runs multi-step AI reasoning to identify the root cause, and auto-populates WB360 tickets with plain-English RCA summaries and recommended next steps, currently live for Ericsson sites and rolling out to Samsung.
- Enabled business users to self-serve decommissioning milestone reporting as measured by 70% reduction in analyst turnaround time, by building a natural language to SQL talk-to-data agent (Qverse) that translates plain-English queries into structured database queries and surfaces real-time progress across non-directional wirecenter consolidation targets.
- Improved cross-system root cause reasoning as measured by 40% reduction in investigation time, by engineering a telecom knowledge graph in Neo4j that maps relationships between network nodes, alarm types, and known failure patterns, enabling hybrid Graph RAG that combines Cypher multi-hop traversal with dense vector retrieval for complex, multi-signal RCA.
- Boosted RAG answer precision by 35% across all deployed agents, by layering Elastic vector search with Cohere reranking, dynamic query expansion, and schema-enforced JSON guardrails that prevent hallucinated structured outputs at inference time.
- Built an agent evaluation harness measuring hallucination rate, tool call accuracy, and final answer quality across 500+ synthetic test cases per release cycle, reducing regression-causing deployments by 50% as measured by production incident logs.
- Established full-stack LLM observability as measured by 30% reduction in debugging cycles, by wiring LangFuse distributed tracing with per-step confidence scoring, token-level cost attribution, and LangSmith regression suites tied to W&B experiment dashboards.

MLOps Engineer (Contract) — Apr 2025 – Aug 2025
Apple Inc., Data Platform Efficiency — *Dallas, TX*

- Engineered a Kubernetes diagnostic system reducing infrastructure triage time by 60% as measured by incident resolution logs, by constructing an MCP-based debugger using LangGraph orchestration and gRPC streaming for real-time pod telemetry.
- Optimized LLM inference latency by 35% as measured by GPU benchmarks, by performing LoRA fine-tuning of Qwen 1.5B on 12K synthetic telemetry samples using Axolotl and deploying via NVIDIA Triton with dynamic batching.
- Improved fine-tuning data quality by generating 12K JSONL training examples via structured prompting, reducing manual annotation cycles by 50% and improving diagnostic output specificity.

- Delivered $1.5M annual cloud savings as measured by cost dashboards, by building Spark pipelines ingesting 5M+ daily GPU metrics into TimescaleDB and surfacing anomalies through automated threshold alerting.
- Maximized GPU utilization by 60% as measured by idle compute reduction, by implementing Run:AI fractional GPU scheduling across shared Ray workloads with priority-based preemption policies.
- Produced $17M monthly cloud cost transparency by generating SKU-level Spark dashboards with automated week-over-week variance forecasting for infrastructure leadership.

Software & AI Engineer May 2024 – Apr 2025
Eminent Services Corporation *Frederick, MD*

- Improved system scalability by 35% and cut maintenance effort by 40%, by migrating a legacy VB6 monolith to a modular MERN stack with Azure DevOps CI/CD and automated regression testing.
- Reduced average API response time by 45% as measured by load test benchmarks, by redesigning core REST endpoints with Node.js, Redis caching, and query optimization across a MongoDB backend.
- Cut manual QA effort by 55% as measured by test cycle duration, by integrating an LLM-assisted test generation pipeline using GPT-4o that produced Playwright regression suites directly from user story descriptions.

ML Research Assistant Jan 2024 – May 2024
University at Buffalo *Buffalo, NY*

- Reduced GPT model energy usage by 20% as measured by inference benchmarks, by applying structured pruning, INT8 quantization, and knowledge distillation to compress transformer architectures for edge deployment.
- Improved cross-domain transfer performance by 12% as measured by zero-shot evaluation benchmarks, by implementing LoRA parameter-efficient fine-tuning on domain-specific scientific corpora using HuggingFace Transformers.

Solutions Engineer Aug 2022 – Aug 2023
Swym Corporation *Bangalore, India*

- Improved wishlist-driven purchase conversion by 28% as measured by A/B test lift across 50+ merchant cohorts, by engineering a neural collaborative filtering recommendation engine trained on 10M+ behavioral events using PyTorch, served via FastAPI on GCP.
- Reduced customer churn by 22% as measured by 90-day retention metrics, by developing an XGBoost propensity model on RFM features with real-time scoring integrated into the merchant analytics dashboard.
- Accelerated merchant analytics reporting 3x as measured by pipeline SLAs, by building Airflow-orchestrated ETL pipelines ingesting Shopify and BigCommerce event streams into a Snowflake warehouse with dbt transformations.
- Improved personalization experiment throughput by 35% as measured by experiment coverage logs, by designing a multi-armed bandit A/B testing framework using Thompson Sampling for real-time offer and recommendation selection.
- Reduced model retraining overhead by 30% as measured by MLflow experiment tracking logs, by building an automated retraining pipeline triggered on data drift signals with versioned model registry and staged rollout via canary deployments.

Projects

Prism-mem | *Python, SQLite, sentence-transformers, MCP, LLM, PyPI* Live | GitHub

- Built an open-source PyPI CLI that extracts project context from Claude Code transcripts and git diffs into semantic triples stored in SQLite with local embeddings, achieving 35x compression (411K bytes condensed to 11.7K characters of queryable knowledge).
- Auto-detects 45.7% of facts as outdated across 416 cross-session connections and regenerates CLAUDE.md, .cursorrules, and AGENTS.md from top-scoring triples; exposes the knowledge base via MCP servers for Claude Code, Cursor, and Codex.

ApplyAI | *LangGraph, Gemini, FastAPI, Next.js, Chrome Extension MV3, Supabase* Live | GitHub

- Built a full-stack job application assistant solo: a Chrome extension with a LangGraph DAG agent that autofills entire application forms (dropdowns, React Select, file uploads), resume-to-job match scoring with keyword breakdown, and an application tracking dashboard. Live on the Chrome Web Store.
- Architected a FastAPI backend with Gemini-powered resume parsing, Supabase vector storage for job-resume similarity scoring, and a Next.js dashboard with real-time application status tracking and KPI analytics.

AtmoFlow | *GCP, Cloud Composer (Airflow), PySpark, BigQuery, Pub/Sub, Looker* GitHub

- Built a real-time weather and air quality data pipeline on GCP using Cloud Functions and Pub/Sub for ingestion, Dataproc PySpark for transformation, BigQuery for warehousing, and Looker dashboards for interactive visualization.
- Implemented schema validation, dead-letter queue handling, and automated anomaly detection on ingested sensor data, reducing data quality incidents by 40% as measured by pipeline monitoring logs.

SecureStack | *AWS EKS, ArgoCD, GitOps, Terraform, DevSecOps, Kubernetes* GitHub

- Designed an end-to-end DevSecOps pipeline with a three-tier architecture on AWS EKS, GitOps-driven deployments via ArgoCD, Terraform-managed infrastructure, and automated security scanning integrated into CI/CD.
- Configured Trivy container scanning, OWASP dependency checks, and OPA policy enforcement as GitHub Actions gates, blocking vulnerable builds from reaching production across all environments.

Education

- **MS, Computer Science**, University at Buffalo GPA: 3.8/4
- **BTech, Computer Science**, Vellore Institute of Technology GPA: 3.9/4